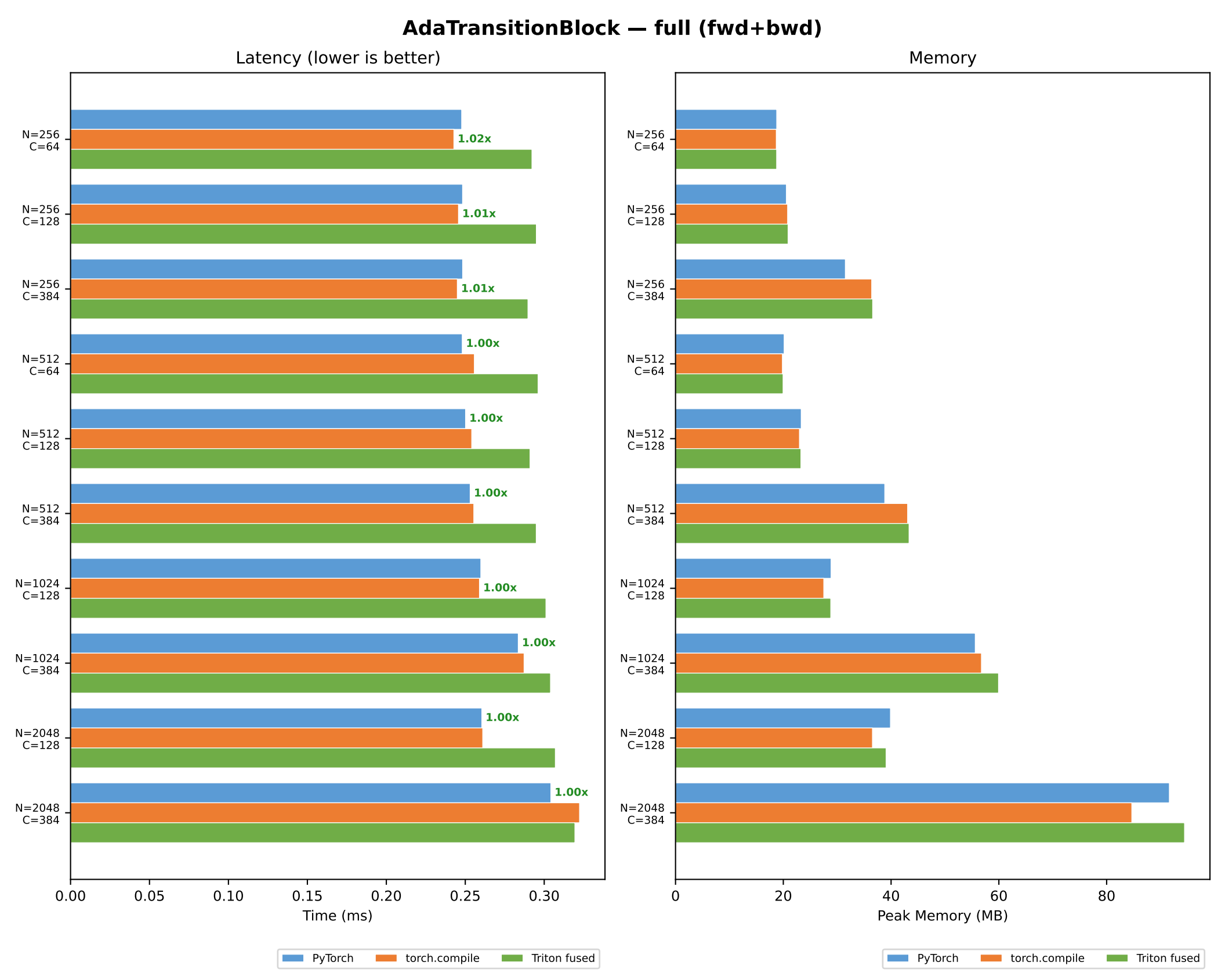
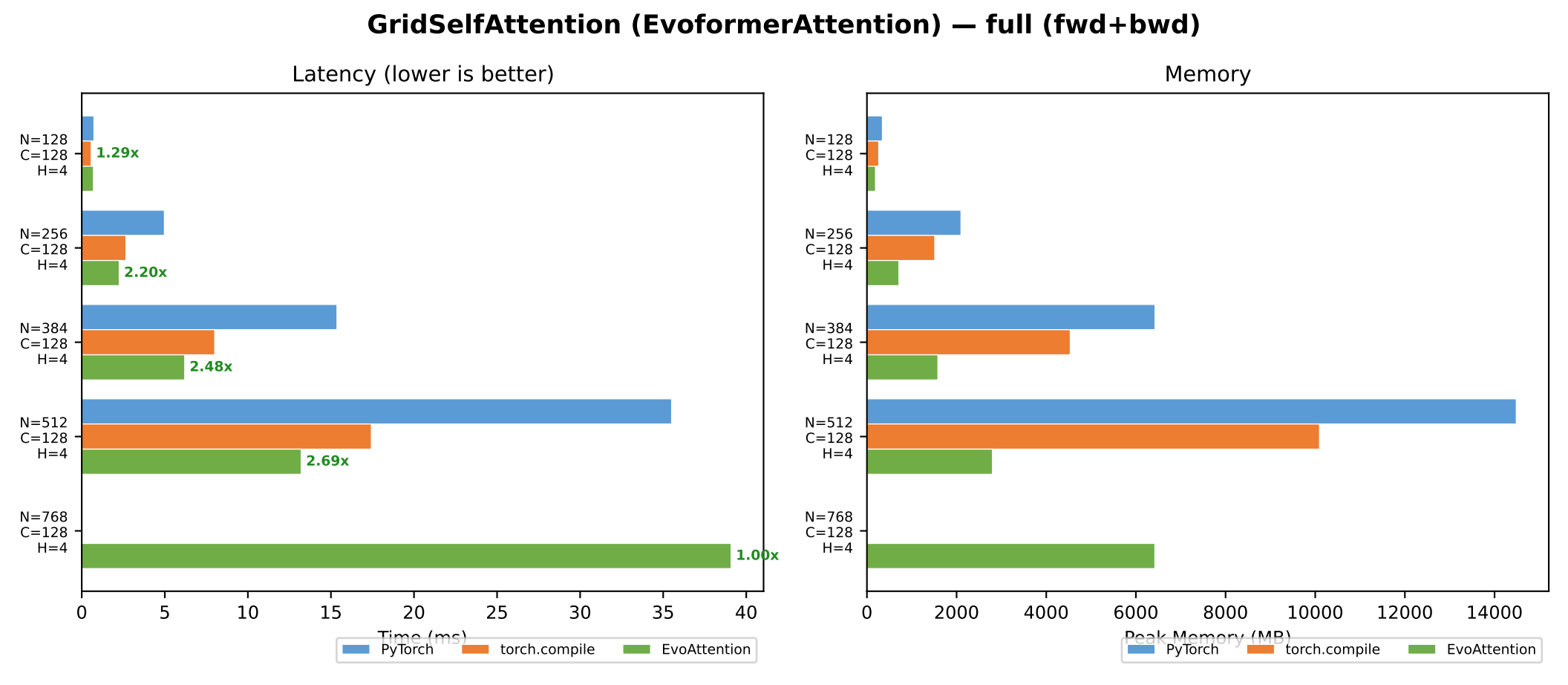
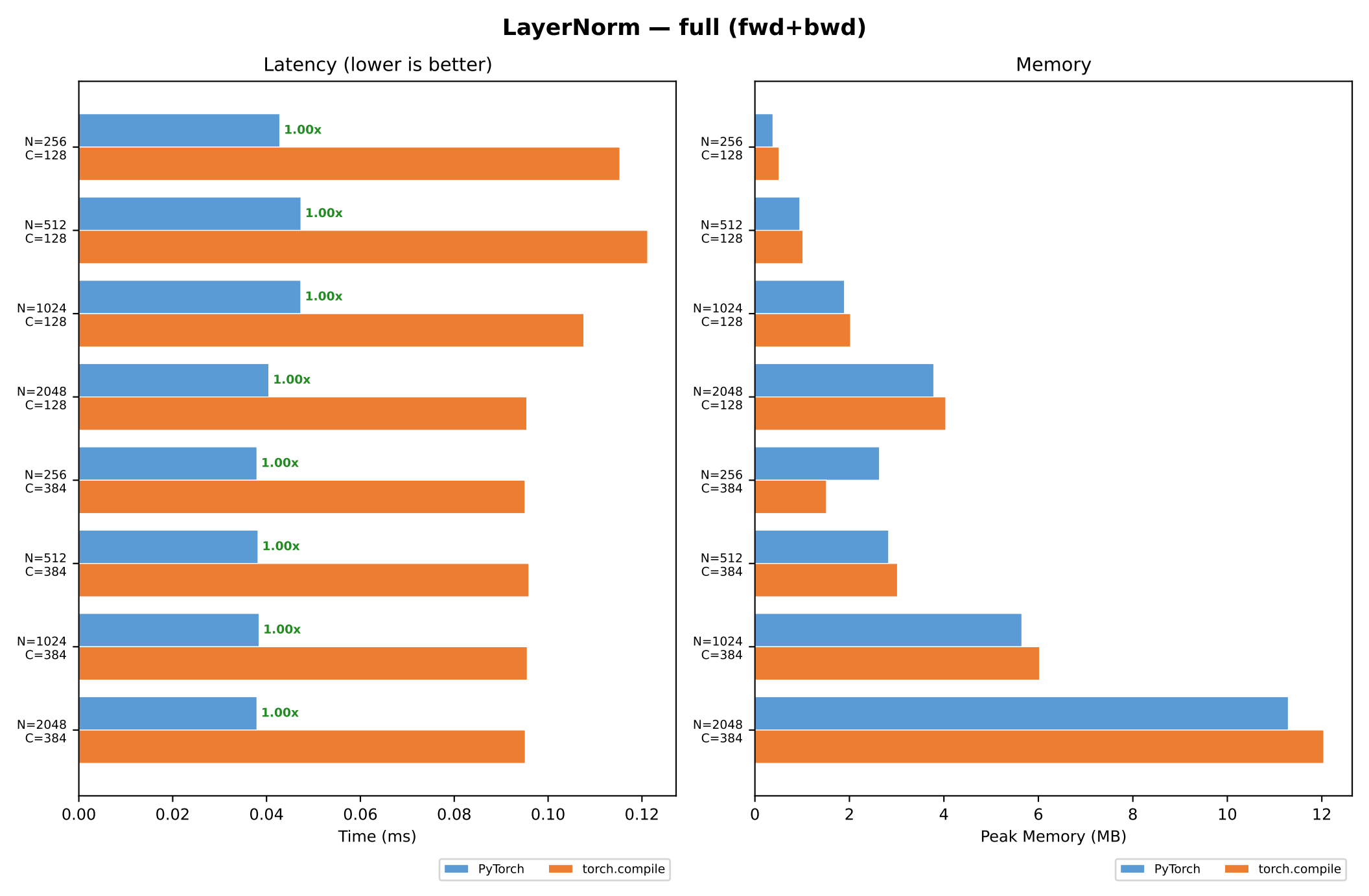
DeepFold3 Kernel Benchmark Report

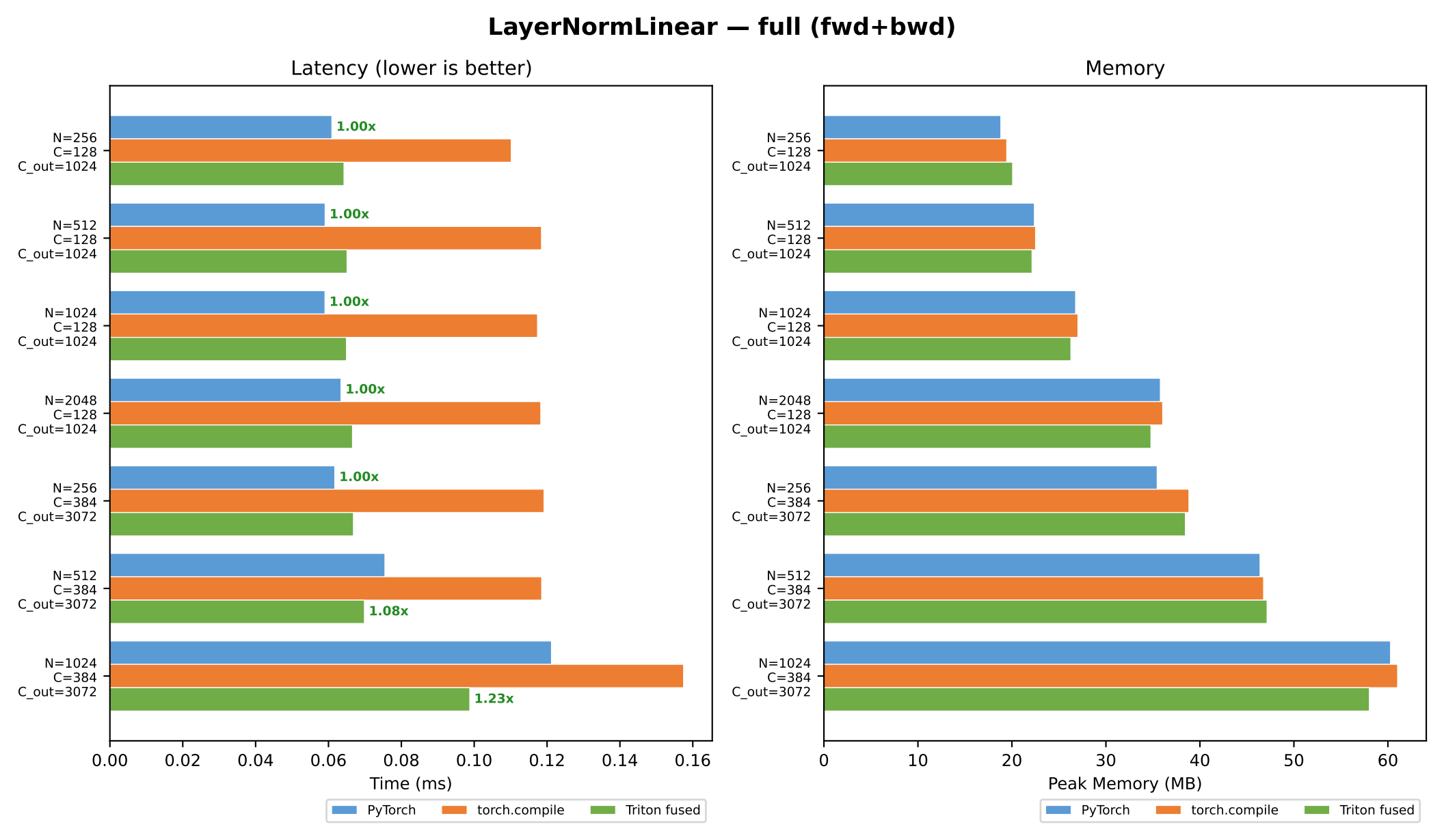
Recommended Optimizations

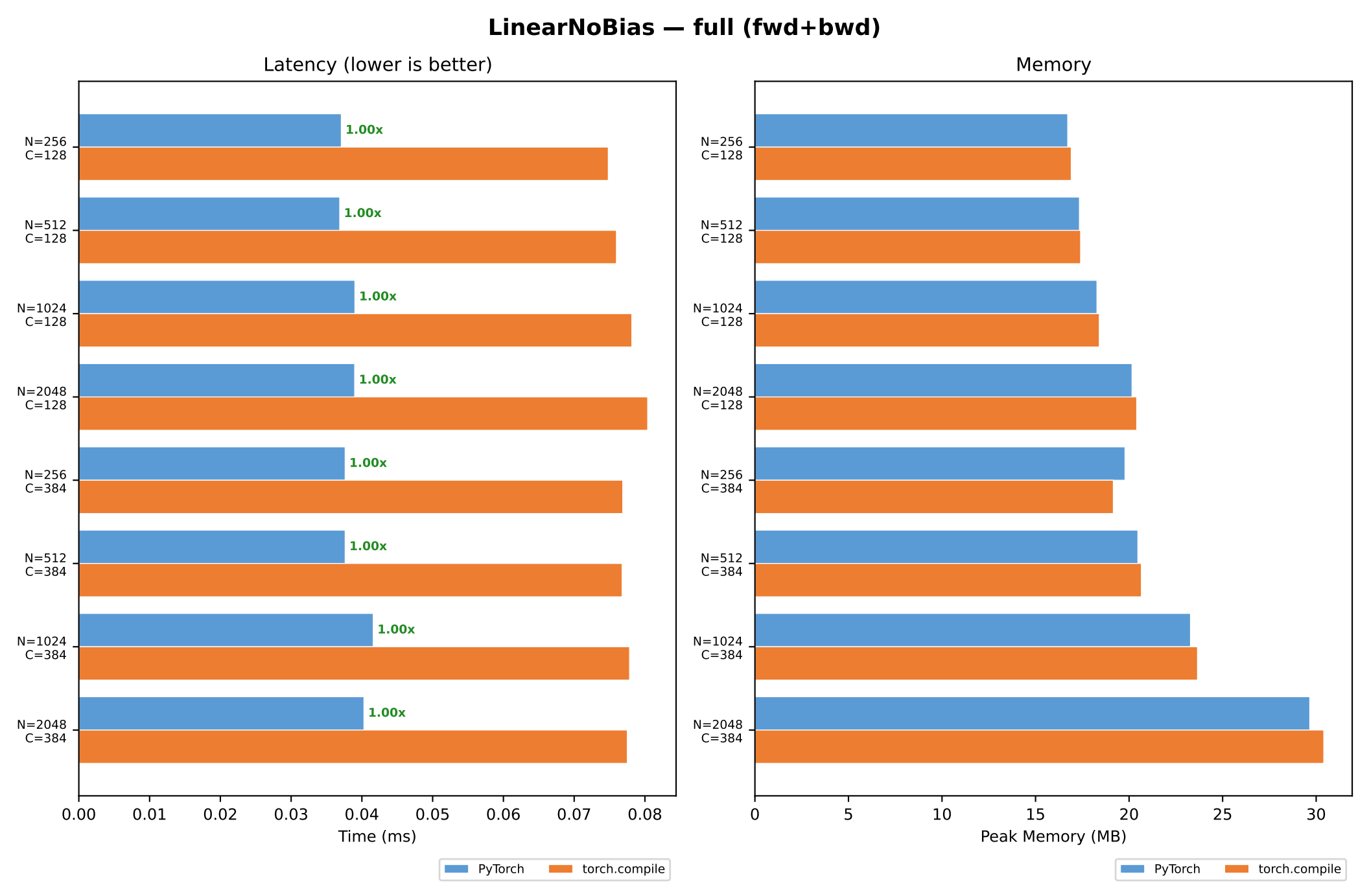
GPU: NVIDIA GeForce RTX 5090 | dtype: bfloat16 | 2026-03-09

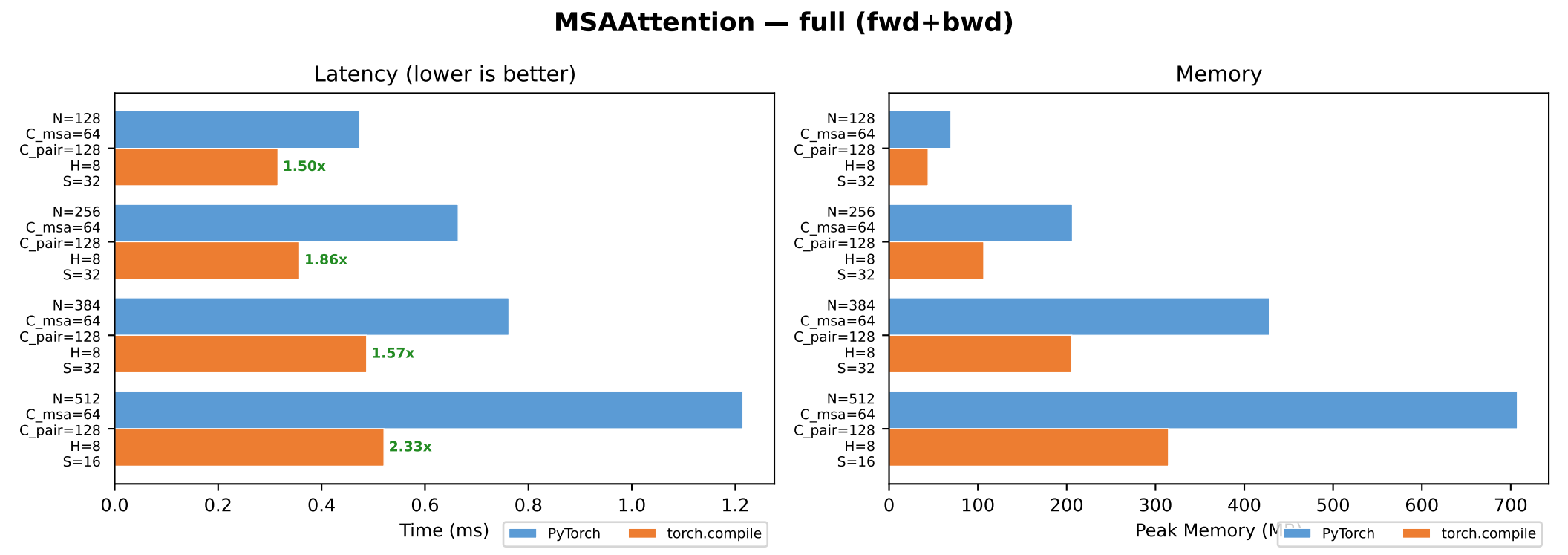

AdaTransitionBlock — full (fwd+bwd)
Latency (lower is better)
Memory
N=256 C=64
N=256 C=128
N=256 C=384
N=512 C=64
N=512 C=128
N=512 C=384
N=1024 C=128
N=1024 C=384
N=2048 C=128
N=2048 C=384
1.02x
1.01x
1.01x
1.00x
1.00x
1.00x
1.00x
1.00x
1.00x
1.00x
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Time (ms)
0
20
40
60
80
Peak Memory (MB)
PyTorch
torch.compile
Triton fused

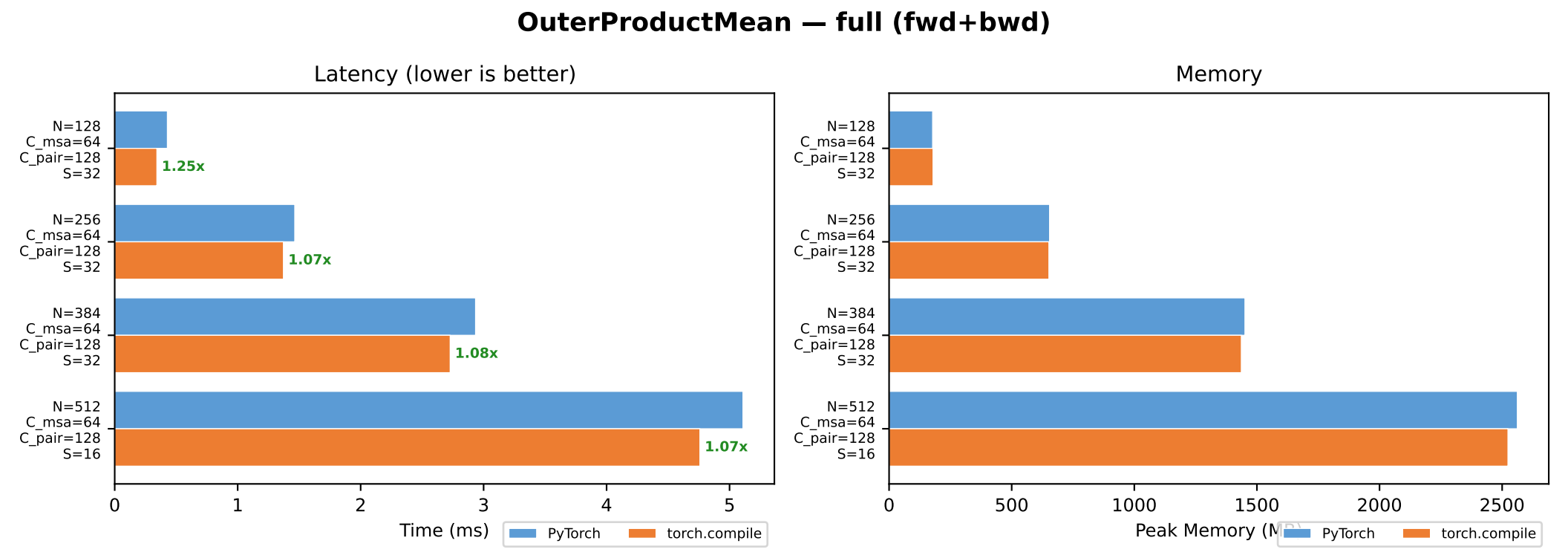

GridSelfAttention (EvoformerAttention) — full (fwd+bwd)
Latency (lower is better)
N=128
C=128
H=4
1.29x
N=256
C=128
H=4
2.20x
N=384
C=128
H=4
2.48x
N=512
C=128
H=4
2.69x
N=768
C=128
H=4
1.00x
0
5
10
15
20
25
30
35
40
Time (ms)
PyTorch
torch.compile
EvoAttention
Memory
N=128
C=128
H=4
N=256
C=128
H=4
N=384
C=128
H=4
N=512
C=128
H=4
N=768
C=128
H=4
0
2000
4000
6000
8000
10000
12000
14000
Peak Memory (MB)
PyTorch
torch.compile
EvoAttention

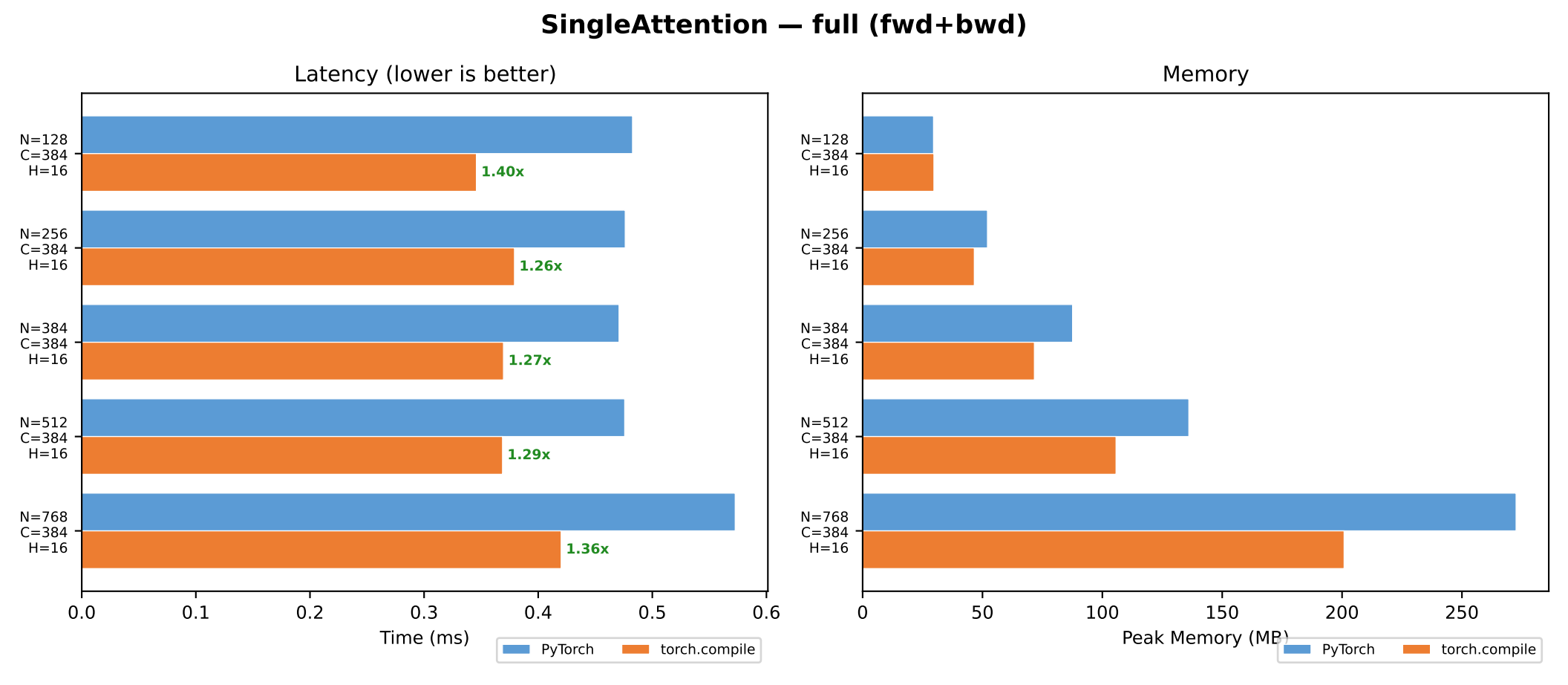

LayerNorm — full (fwd+bwd)
Latency (lower is better)
Memory
N=256 C=128
N=512 C=128
N=1024 C=128
N=2048 C=128
N=256 C=384
N=512 C=384
N=1024 C=384
N=2048 C=384
1.00x
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Time (ms)
0
2
4
6
8
10
12
Peak Memory (MB)
PyTorch
torch.compile

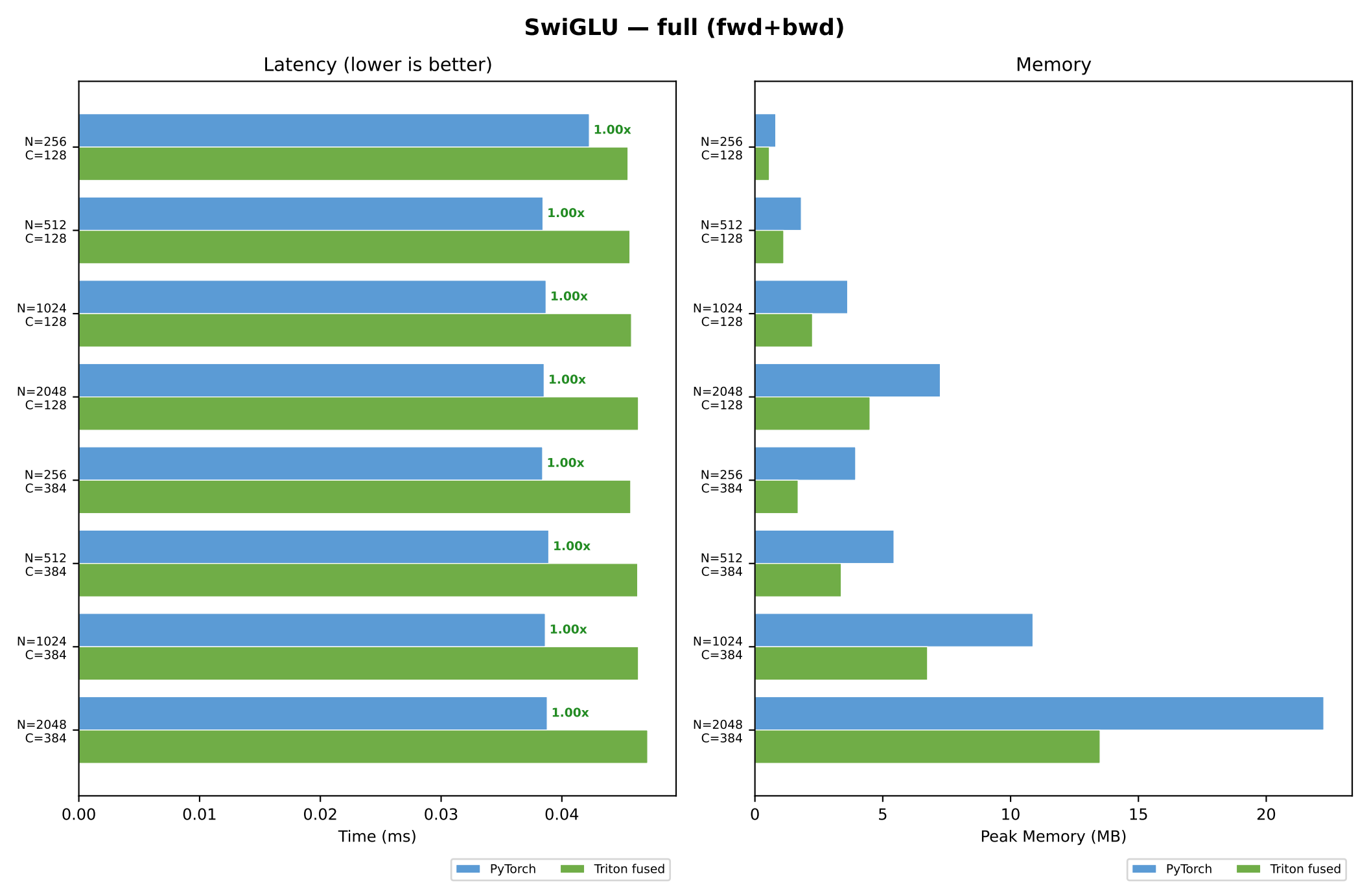

LayerNormLinear — full (fwd+bwd)
Latency (lower is better)
Memory
N=256 C=128 C_out=1024
N=512 C=128 C_out=1024
N=1024 C=128 C_out=1024
N=2048 C=128 C_out=1024
N=256 C=384 C_out=3072
N=512 C=384 C_out=3072
N=1024 C=384 C_out=3072
1.00x
1.00x
1.00x
1.00x
1.00x
1.08x
1.23x
0.00 0.02 0.04 0.06 0.08 0.10 0.12 0.14 0.16
Time (ms)
0 10 20 30 40 50 60
Peak Memory (MB)
PyTorch
torch.compile
Triton fused

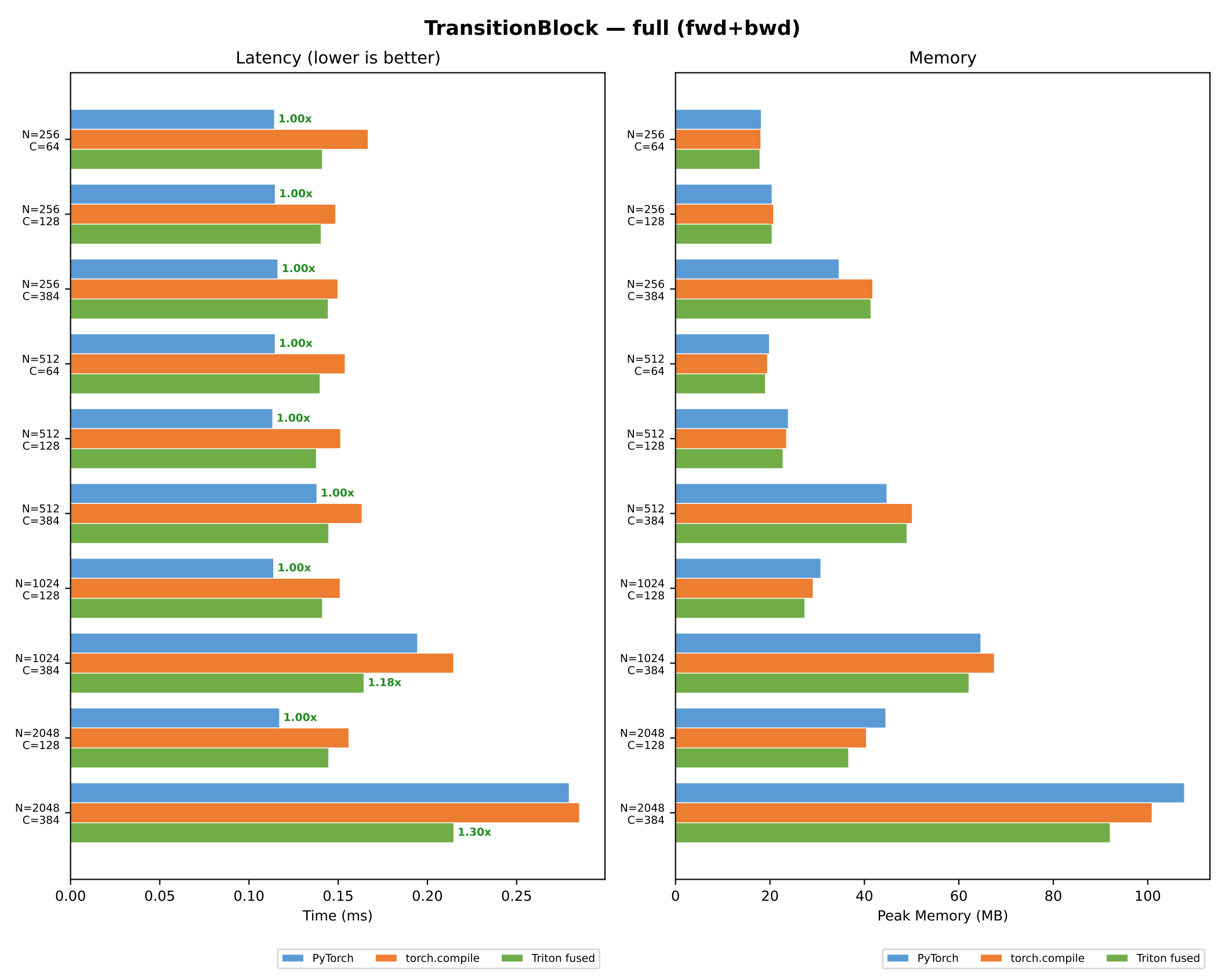

LinearNoBias — full (fwd+bwd)
Latency (lower is better)
Memory
N=256 C=128
N=512 C=128
N=1024 C=128
N=2048 C=128
N=256 C=384
N=512 C=384
N=1024 C=384
N=2048 C=384
1.00x
0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08
Time (ms)
0 5 10 15 20 25 30
Peak Memory (MB)
PyTorch
torch.compile

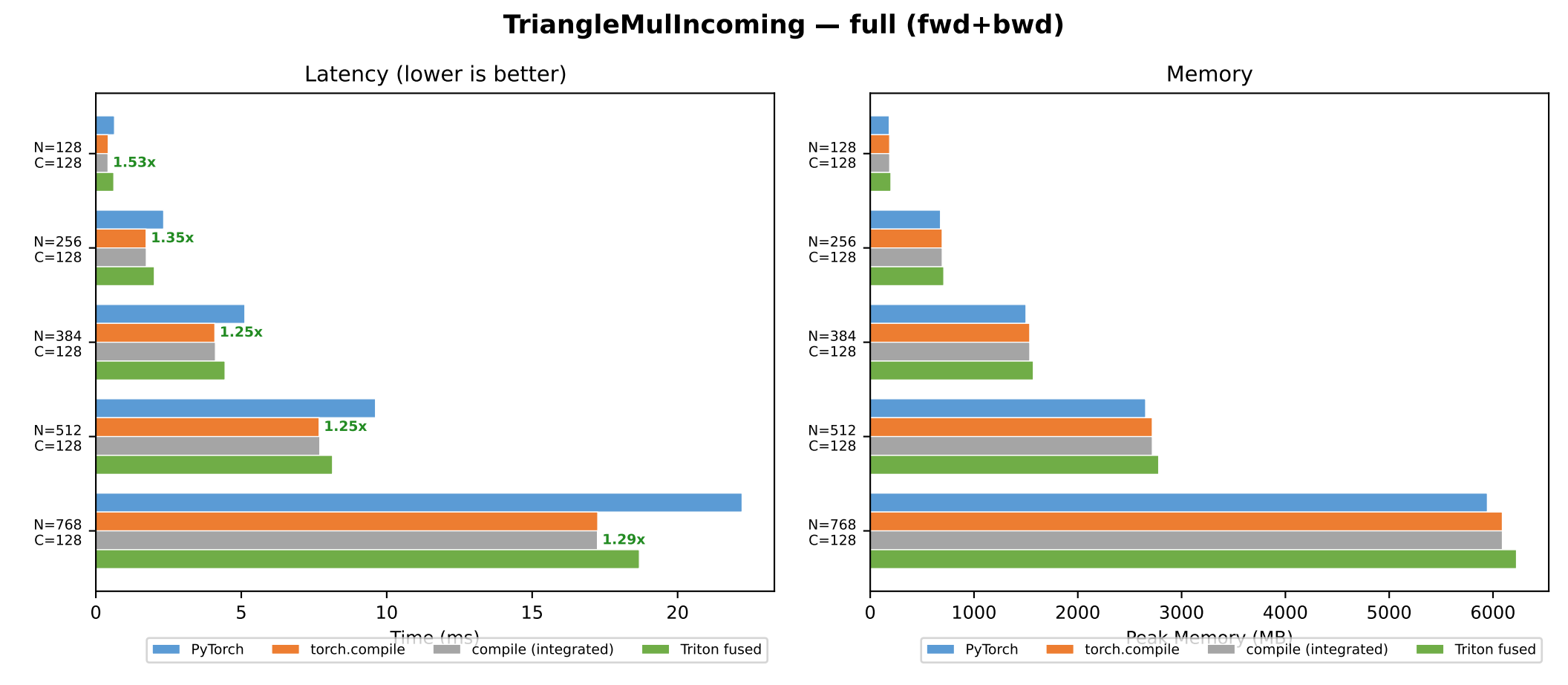

MSAAttention — full (fwd+bwd)
Latency (lower is better)
N=128
C_msa=64
C_pair=128
H=8
S=32
1.50x
N=256
C_msa=64
C_pair=128
H=8
S=32
1.86x
N=384
C_msa=64
C_pair=128
H=8
S=32
1.57x
N=512
C_msa=64
C_pair=128
H=8
S=16
2.33x
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Time (ms)
PyTorch
torch.compile
Memory
N=128
C_msa=64
C_pair=128
H=8
S=32
N=256
C_msa=64
C_pair=128
H=8
S=32
N=384
C_msa=64
C_pair=128
H=8
S=32
N=512
C_msa=64
C_pair=128
H=8
S=16
0
100
200
300
400
500
600
700
Peak Memory (M
PyTorch
torch.compile

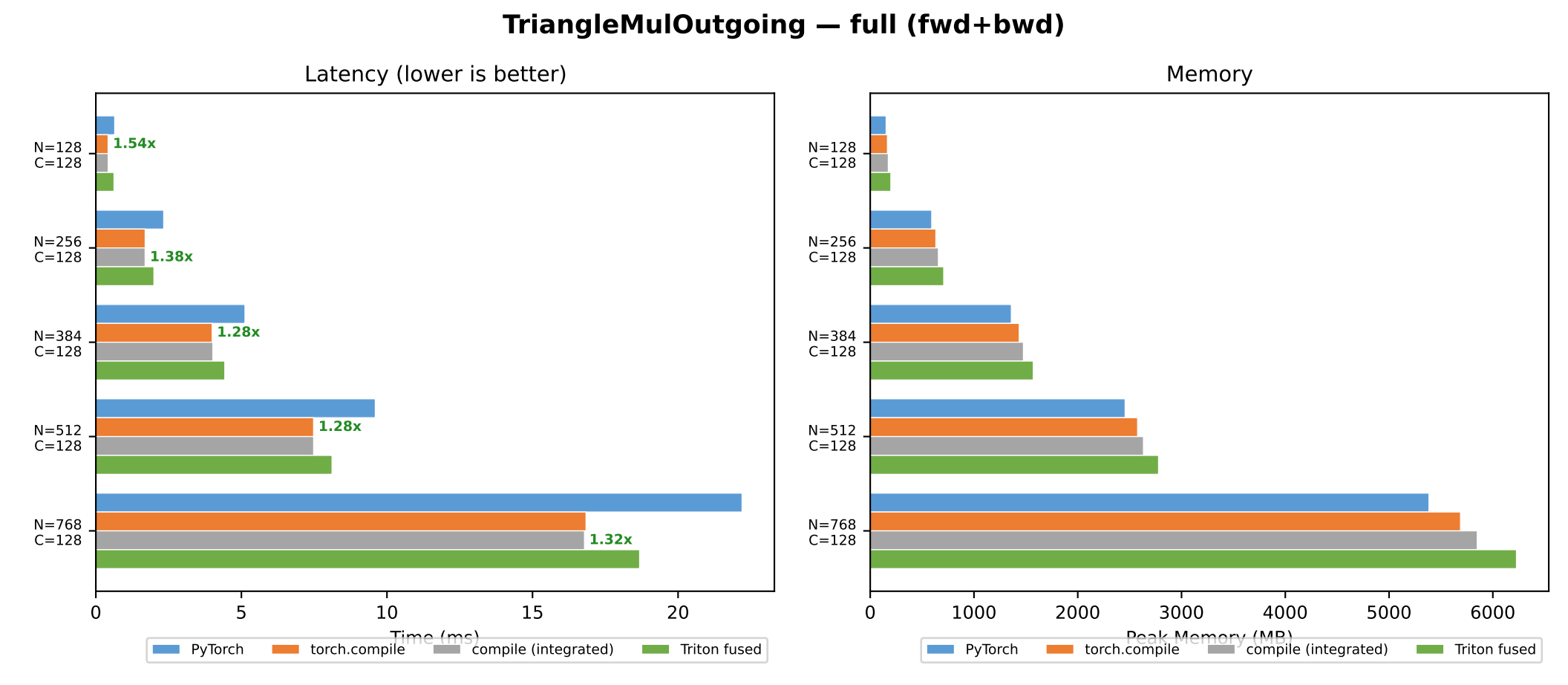

OuterProductMean — full (fwd+bwd)
Latency (lower is better)
N=128
C_msa=64
C_pair=128
S=32
1.25x
N=256
C_msa=64
C_pair=128
S=32
1.07x
N=384
C_msa=64
C_pair=128
S=32
1.08x
N=512
C_msa=64
C_pair=128
S=16
1.07x
0
1
2
3
4
5
Time (ms)
PyTorch
torch.compile
Memory
N=128
C_msa=64
C_pair=128
S=32
N=256
C_msa=64
C_pair=128
S=32
N=384
C_msa=64
C_pair=128
S=32
N=512
C_msa=64
C_pair=128
S=16
0
500
1000
1500
2000
2500
Peak Memory (M
PyTorch
torch.compile


SingleAttention — full (fwd+bwd)
Latency (lower is better)
N=128
C=384
H=16
1.40x
N=256
C=384
H=16
1.26x
N=384
C=384
H=16
1.27x
N=512
C=384
H=16
1.29x
N=768
C=384
H=16
1.36x
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Time (ms)
PyTorch
torch.compile
Memory
N=128
C=384
H=16
N=256
C=384
H=16
N=384
C=384
H=16
N=512
C=384
H=16
N=768
C=384
H=16
0
50
100
150
200
250
Peak Memory (MB)
PyTorch
torch.compile


SwiGLU — full (fwd+bwd)
Latency (lower is better)
Memory
N=256 C=128
N=512 C=128
N=1024 C=128
N=2048 C=128
N=256 C=384
N=512 C=384
N=1024 C=384
N=2048 C=384
1.00x
0.00
0.01
0.02
0.03
0.04
Time (ms)
0
5
10
15
20
Peak Memory (MB)
PyTorch
Triton fused


TransitionBlock — full (fwd+bwd)
Latency (lower is better)
Memory
N=256 C=64
N=256 C=128
N=256 C=384
N=512 C=64
N=512 C=128
N=512 C=384
N=1024 C=128
N=1024 C=384
N=2048 C=128
N=2048 C=384
1.00x
1.00x
1.00x
1.00x
1.00x
1.00x
1.00x
1.18x
1.00x
1.30x
0.00
0.05
0.10
0.15
0.20
0.25
Time (ms)
0
20
40
60
80
100
Peak Memory (MB)
PyTorch
torch.compile
Triton fused


TriangleMulIncoming — full (fwd+bwd)
Latency (lower is better)
Memory
N=128
C=128
N=256
C=128
N=384
C=128
N=512
C=128
N=768
C=128
1.53x
1.35x
1.25x
1.25x
1.29x
0
5
10
15
20
Time (ms)
0
1000
2000
3000
4000
5000
6000
Peak Memory (MB)
PyTorch
torch.compile
compile (integrated)
Triton fused


TriangleMulOutgoing — full (fwd+bwd)
Latency (lower is better)
Memory
N=128
C=128
N=256
C=128
N=384
C=128
N=512
C=128
N=768
C=128
1.54x
1.38x
1.28x
1.28x
1.32x
0
5
10
15
20
Time (ms)
0
1000
2000
3000
4000
5000
6000
Peak Memory (MB)
PyTorch
torch.compile
compile (integrated)
Triton fused

Recommendation Summary

Operation	Shape	Recommended	Time (ms)	Speedup
AdaTransitionBlock	N=256 / C=64	torch.compile	0.243	1.02x
AdaTransitionBlock	N=256 / C=128	torch.compile	0.246	1.01x
AdaTransitionBlock	N=256 / C=384	torch.compile	0.245	1.01x
AdaTransitionBlock	N=512 / C=64	PyTorch	0.248	baseline
AdaTransitionBlock	N=512 / C=128	PyTorch	0.250	baseline
AdaTransitionBlock	N=512 / C=384	PyTorch	0.253	baseline
AdaTransitionBlock	N=1024 / C=128	torch.compile	0.259	baseline
AdaTransitionBlock	N=1024 / C=384	PyTorch	0.284	baseline
AdaTransitionBlock	N=2048 / C=128	PyTorch	0.260	baseline
AdaTransitionBlock	N=2048 / C=384	PyTorch	0.304	baseline
GridSelfAttention	N=128 / C=128 / H=4	torch.compile	0.568	1.29x
GridSelfAttention	N=256 / C=128 / H=4	EvoAttention	2.256	2.20x
GridSelfAttention	N=384 / C=128 / H=4	EvoAttention	6.199	2.48x
GridSelfAttention	N=512 / C=128 / H=4	EvoAttention	13.212	2.69x
GridSelfAttention	N=768 / C=128 / H=4	EvoAttention	39.087	baseline
LayerNorm	N=256 / C=128	PyTorch	0.043	baseline
LayerNorm	N=512 / C=128	PyTorch	0.047	baseline
LayerNorm	N=1024 / C=128	PyTorch	0.047	baseline
LayerNorm	N=2048 / C=128	PyTorch	0.040	baseline
LayerNorm	N=256 / C=384	PyTorch	0.038	baseline
LayerNorm	N=512 / C=384	PyTorch	0.038	baseline
LayerNorm	N=1024 / C=384	PyTorch	0.038	baseline
LayerNorm	N=2048 / C=384	PyTorch	0.038	baseline
LayerNormLinear	N=256 / C=128 / C_out=1024	PyTorch	0.061	baseline
LayerNormLinear	N=512 / C=128 / C_out=1024	PyTorch	0.059	baseline
LayerNormLinear	N=1024 / C=128 / C_out=1024	PyTorch	0.059	baseline
LayerNormLinear	N=2048 / C=128 / C_out=1024	PyTorch	0.063	baseline
LayerNormLinear	N=256 / C=384 / C_out=3072	PyTorch	0.062	baseline
LayerNormLinear	N=512 / C=384 / C_out=3072	Triton fused	0.070	1.08x
LayerNormLinear	N=1024 / C=384 / C_out=3072	Triton fused	0.099	1.23x
LinearNoBias	N=256 / C=128	PyTorch	0.037	baseline
LinearNoBias	N=512 / C=128	PyTorch	0.037	baseline
LinearNoBias	N=1024 / C=128	PyTorch	0.039	baseline
LinearNoBias	N=2048 / C=128	PyTorch	0.039	baseline
LinearNoBias	N=256 / C=384	PyTorch	0.038	baseline

Operation	Shape	Recommended	Time (ms)	Speedup
LinearNoBias	N=512 / C=384	PyTorch	0.038	baseline
LinearNoBias	N=1024 / C=384	PyTorch	0.042	baseline
LinearNoBias	N=2048 / C=384	PyTorch	0.040	baseline
MSAAttention	N=128 / C_msa=64 / C_pair=128 / H=8 / S=32	torch.compile	0.316	1.50x
MSAAttention	N=256 / C_msa=64 / C_pair=128 / H=8 / S=32	torch.compile	0.357	1.86x
MSAAttention	N=384 / C_msa=64 / C_pair=128 / H=8 / S=32	torch.compile	0.487	1.57x
MSAAttention	N=512 / C_msa=64 / C_pair=128 / H=8 / S=16	torch.compile	0.521	2.33x
OuterProductMean	N=128 / C_msa=64 / C_pair=128 / S=32	torch.compile	0.343	1.25x
OuterProductMean	N=256 / C_msa=64 / C_pair=128 / S=32	torch.compile	1.372	1.07x
OuterProductMean	N=384 / C_msa=64 / C_pair=128 / S=32	torch.compile	2.728	1.08x
OuterProductMean	N=512 / C_msa=64 / C_pair=128 / S=16	torch.compile	4.759	1.07x
SingleAttention	N=128 / C=384 / H=16	torch.compile	0.346	1.40x
SingleAttention	N=256 / C=384 / H=16	torch.compile	0.379	1.26x
SingleAttention	N=384 / C=384 / H=16	torch.compile	0.370	1.27x
SingleAttention	N=512 / C=384 / H=16	torch.compile	0.369	1.29x
SingleAttention	N=768 / C=384 / H=16	torch.compile	0.420	1.36x
SwiGLU	N=256 / C=128	PyTorch	0.042	baseline
SwiGLU	N=512 / C=128	PyTorch	0.038	baseline
SwiGLU	N=1024 / C=128	PyTorch	0.039	baseline
SwiGLU	N=2048 / C=128	PyTorch	0.039	baseline
SwiGLU	N=256 / C=384	PyTorch	0.038	baseline
SwiGLU	N=512 / C=384	PyTorch	0.039	baseline
SwiGLU	N=1024 / C=384	PyTorch	0.039	baseline
SwiGLU	N=2048 / C=384	PyTorch	0.039	baseline
TransitionBlock	N=256 / C=64	PyTorch	0.114	baseline
TransitionBlock	N=256 / C=128	PyTorch	0.115	baseline
TransitionBlock	N=256 / C=384	PyTorch	0.116	baseline
TransitionBlock	N=512 / C=64	PyTorch	0.115	baseline
TransitionBlock	N=512 / C=128	PyTorch	0.113	baseline
TransitionBlock	N=512 / C=384	PyTorch	0.138	baseline
TransitionBlock	N=1024 / C=128	PyTorch	0.114	baseline
TransitionBlock	N=1024 / C=384	Triton fused	0.164	1.18x
TransitionBlock	N=2048 / C=128	PyTorch	0.117	baseline
TransitionBlock	N=2048 / C=384	Triton fused	0.215	1.30x
TriangleMulIncoming	N=128 / C=128	compile (integrated)	0.418	1.53x

Operation	Shape	Recommended	Time (ms)	Speedup
TriangleMulIncoming	N=256 / C=128	torch.compile	1.727	1.35x
TriangleMulIncoming	N=384 / C=128	torch.compile	4.090	1.25x
TriangleMulIncoming	N=512 / C=128	torch.compile	7.674	1.25x
TriangleMulIncoming	N=768 / C=128	compile (integrated)	17.243	1.29x
TriangleMulOutgoing	N=128 / C=128	torch.compile	0.420	1.54x
TriangleMulOutgoing	N=256 / C=128	compile (integrated)	1.692	1.38x
TriangleMulOutgoing	N=384 / C=128	torch.compile	3.995	1.28x
TriangleMulOutgoing	N=512 / C=128	torch.compile	7.480	1.28x
TriangleMulOutgoing	N=768 / C=128	compile (integrated)	16.784	1.32x